Take-Home Assignment:
A Learning Mechanism for Manufacturing Operators

Test Duration: 7 days

Objective

Design and demonstrate a simple agentic AI assistant that can learn behavioural patterns of a manufacturing operator over time.

The goal is not to build a production-ready system. We are more interested in your solution design thinking, assumptions, trade-offs, and how you test whether the idea is feasible.

Background

A manufacturing company wants to build an AI assistant for shopfloor operators.

The assistant should gradually learn how each operator works, such as:

- Preferred way of receiving instructions
- Escalation behaviour
- Troubleshooting habits
- Shift-related work patterns
- Learning needs
- Confidence with different machine/process issues

These behavioural patterns cannot be accurately captured from a single conversation. They must be inferred and validated over repeated interactions and operational data over time.

Task 1: Solution Design

Propose an agentic AI architecture for this assistant.

Your design should explain:

1. What behavioural patterns you would capture
2. What data sources you would use
3. What agents/components are needed
4. How the system learns over time
5. How memory is stored, updated, and corrected
6. How the assistant avoids wrong assumptions
7. How the profile is used to personalize future support

You may include architecture diagrams if useful.

Task 2: Simple Demo

Build a simple working demo using any programming language, framework, or technology stack of your choice.

The demo should utilize at least one real Large Language Model (LLM) API. You may use commercial or open-source hosted models.

The demo does not need to use real manufacturing data. Simulated or mock data is acceptable.

The demo should demonstrate how the assistant can:

1. Receive repeated operator interactions or events
2. Extract behavioural signals from interactions
3. Update an operator profile over time
4. Assign confidence levels to inferred patterns
5. Use the evolving profile to personalize future responses

Example Demo Scenario

- Operator A repeatedly requests visual instructions
- Operator A resolves simple machine alarms independently
- Operator A escalates complex issues quickly

The system gradually infers:

- Prefers visual guidance
- Confident with basic troubleshooting
- Requires support for complex fault diagnosis

The demo can take any form.

The focus is on demonstrating the concept, reasoning, and behavioural learning process rather than building a production-grade application.

Please include a brief explanation of:

- Your chosen architecture
- Key design decisions
- Assumptions made
- Known limitations of the prototype

Task 3: Design of Experiment

Assume your proposed solution is expanded into a pilot involving manufacturing operators.

Briefly describe how you would validate whether the assistant is successfully learning useful behavioural patterns and providing meaningful personalization.

Please include:

1. Your proposed hypothesis
2. Key metrics you would track
3. How you would verify that the behavioural profile is accurate
4. At least 3 potential risks, failure modes, or loopholes you would watch for

This section is intended to assess your approach to validating AI systems rather than to produce a detailed experimental plan.

Expected Deliverables

Please submit:

1. A short write-up or slide deck explaining your concept and architecture
2. A simple working demo or prototype utilizing a real LLM API
3. Instructions on how to run the demo
4. A brief experiment validation proposal

Slides are optional but encouraged if they help communicate your idea clearly.